

PREALGEBRA

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Algebra

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1. The Whole Numbers

1.1. Place Value, Names for Numbers, and Tables

The **whole numbers** are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,

The **natural numbers** are 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,

1.1.1. Finding the Place Value of a Digit in a Whole Number

The position of each digit in a number determines its **place value**.

Periods											
Billions			Millions			Thousands			Ones		
Hundred-billions	Ten-billions	Billion	Hundred-millions	Ten-millions	Millions	Hundred-thousands	Ten-thousands	Thousands	Hundreds	Tens	Ones

1.1.2. Writing a Whole Number in Words and in Standard Form

A whole number such as 1,083,664,500 is written in **standard form**. Commas separate the digits into groups of three, starting from the right. Each group of three digits is called a **period**.

Writing a Whole Number in Words

To write a whole number in words, write the number in each period followed by the name of the period. This same procedure can be used to read a whole number.

Example 9,265 is read as “nine thousand, two hundred sixty five.”

We write 1,083,664,500 as “one billion, eighty-three million, six hundred sixty-four thousand, five hundred”.

Writing a Whole Number in Standard Form

To write a whole number in standard form, write the number in each period, followed by a comma.

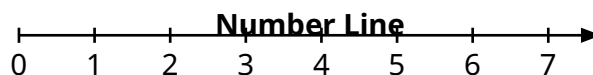
1.1.3. Writing a Whole Number in Expanded Form

The place value of a digit can be used to write a number in expanded form. The **expanded form** of a number shows each digit of the number with its place value.

Example 5672 is written in expanded form as

$$5672 = 5000 + 600 + 70 + 2$$

We can visualize whole numbers by points on a line. The line below is called a **number line**. This number line has equally spaced marks for each whole number. The arrow to the right means that the whole numbers continue indefinitely.



1.1.4. Tables

Tables are often used to organize and display facts that contain numbers.

Number of notebooks	Cost per notebook	Total cost
1	\$3	\$3
2	\$3	\$6
3	\$3	\$9
4	\$3	\$12
5	\$3	\$15

1.2. Adding and Subtracting Whole Numbers, and Perimeter

1.2.1. Adding Whole Numbers

The **sum** of two numbers is their **total**. The numbers being added are called **addends**, and the process of finding their sum is called **addition**.

$$\begin{array}{ccccc}
 2 & + & 4 & = & 6 \\
 \uparrow & & \uparrow & & \uparrow \\
 \text{addend} & & \text{addend} & & \text{sum}
 \end{array}$$

To add whole numbers, we add the digits in the ones place, then the tens place, then the hundreds place, and so on.

Example Line up numbers vertically so that the place values correspond. Then add digits in corresponding place values, starting with the ones place.

$$\begin{array}{r}
 2236 \\
 + \quad 160 \\
 \hline
 1396
 \end{array}$$

■

When the sum of digits in corresponding place values is more than 9, **carrying** is necessary.

Example To add $365 + 89$, add the ones-place digits first.

$$\begin{array}{r}
 1 \\
 365 \\
 + \quad 89 \\
 \hline
 4
 \end{array}$$

5 ones + 9 ones = 14 ones or 1 ten + 4 ones

4 Write the 4 ones in the ones place and carry the 1 ten to the tens place.

Next, add the tens-place digits.

$$\begin{array}{r}
 1 \quad 1 \\
 365 \\
 + \quad 89 \\
 \hline
 5 \quad 4
 \end{array}$$

1 ten + 6 tens + 8 tens = 15 tens or 1 hundred + 5 tens

5 4 Write the 5 tens in the tens place and carry the 1 hundred to the hundreds place.

Next, add the hundreds-place digits.

$$\begin{array}{r}
 1 \quad 1 \\
 365 \\
 + \quad 89 \\
 \hline
 4 \quad 5 \quad 4
 \end{array}$$

1 hundred + 3 hundreds = 4 hundreds

4 5 4 Write the 4 hundreds in the hundreds place.

■

Addition Property of 0

The sum of 0 and any number is that number. For example,

$$7 + 0 = 7$$

$$0 + 7 = 7$$

Commutative Property of Addition

Changing the **order** of two addends does not change their sum. For example,

$$2 + 3 = 5 \quad \text{and} \quad 3 + 2 = 5$$

Associative Property of Addition

Changing the **grouping** of addends does not change their sum. For example,

$$3 + (5 + 7) = 3 + 12 = 15 \quad \text{and} \quad (3 + 5) + 7 = 8 + 7 = 15$$

When adding several numbers, it is often helpful to look for two or three numbers whose sum is 10, 20, and so on.

1.2.2. Subtracting Whole Numbers

Subtraction is finding the **difference** of two numbers.

$$\begin{array}{ccccccc} 8 & - & 5 & = & 3 \\ \uparrow & & \uparrow & & \uparrow \\ \text{minuend} & & \text{subtrahend} & & \text{difference} \end{array}$$

Subtraction is defined in terms of addition.

$$8 - 3 = 5 \quad \text{because} \quad 5 + 3 = 8$$

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